

Agro-Smart

An IoT-Based Smart Agriculture Monitoring System
with AI-Assisted Precision Farming and Cloud
Analytics for Location-Specific Decision Support

- KAHNDSE25.1F-005
- KAHNDSE25.1F-007
- KAHNDSE25.1F-009
- KAHNDSE25.1F-032



Introduction to AgroSmart

- Agriculture depends heavily on environmental conditions
- Traditional farming lacks real-time monitoring and data-driven decisions
- Water wastage and inefficient crop planning are common issues
- IoT, data analytics, and AI can modernize agriculture
- This project proposes a complete smart agriculture solution



Project Motivation

- Farmers need:
 - Real-time field data
 - Better irrigation decisions
 - Climate-aware guidance
 - Many existing systems focus only on automation
 - This project focuses on:
 - Monitoring first
 - Intelligent insights second
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Project Scope

- Develop a Smart Agriculture Monitoring System
- Integrate a Smart Precision Agriculture System as a sub-part
- Initial deployment for a specific geographic area
 - Example: Kandy, Sri Lanka
- System is scalable for future expansion
- Additional features can be added based on time and cost



System Overview

Main System:

Smart Agriculture Monitoring System

- Temperature monitoring
- Humidity monitoring
- Soil moisture monitoring
- Smart irrigation monitoring
- Cloud data storage
- Web dashboard & mobile app

Sub-System:

Smart Precision Agriculture System

- Data analysis
- AI/ML-based decision support
- Intelligent recommendations



Role of IoT

- Sensors collect real-time agricultural data
- ESP32 microcontroller processes data
- Wireless communication via Wi-Fi
- Data sent to Firebase cloud
- Enables remote monitoring and control

IoT Components:

- DHT22 (Temperature & Humidity)
 - Soil Moisture Sensor
 - ESP32
 - Firebase Cloud
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Data Variance & Types of Data

Collected Data Types:

- Temperature (°C)
- Humidity (%)
- Soil moisture (%)
- Irrigation status (ON/OFF)
- Irrigation duration
- Timestamp (date & time)

Nature of Data:

- Numerical
- Boolean
- Time-series



Data Collection Methodology

Parameter	Sensor / Source
Temperature	DHT22
Soil Moisture	Capacitive Sensor
Humidity	DHT22
Pump Status	ESP32
Time	System Timestamp

- Data collected periodically
 - Stored in Firebase
 - Used for monitoring and analysis
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Cloud Platform & Visualization

- Firebase Realtime Database
 - Real-time data synchronization
 - Secure cloud storage
 - Web dashboard:
 - Live sensor values
 - Trends and charts
 - Mobile application:
 - Remote monitoring
 - Alerts & insights
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Precision Agriculture System

- Uses data collected by monitoring system
- No separate sensors
- Includes:
 - Data analysis module
 - AI/ML models
- Works as an intelligent decision layer



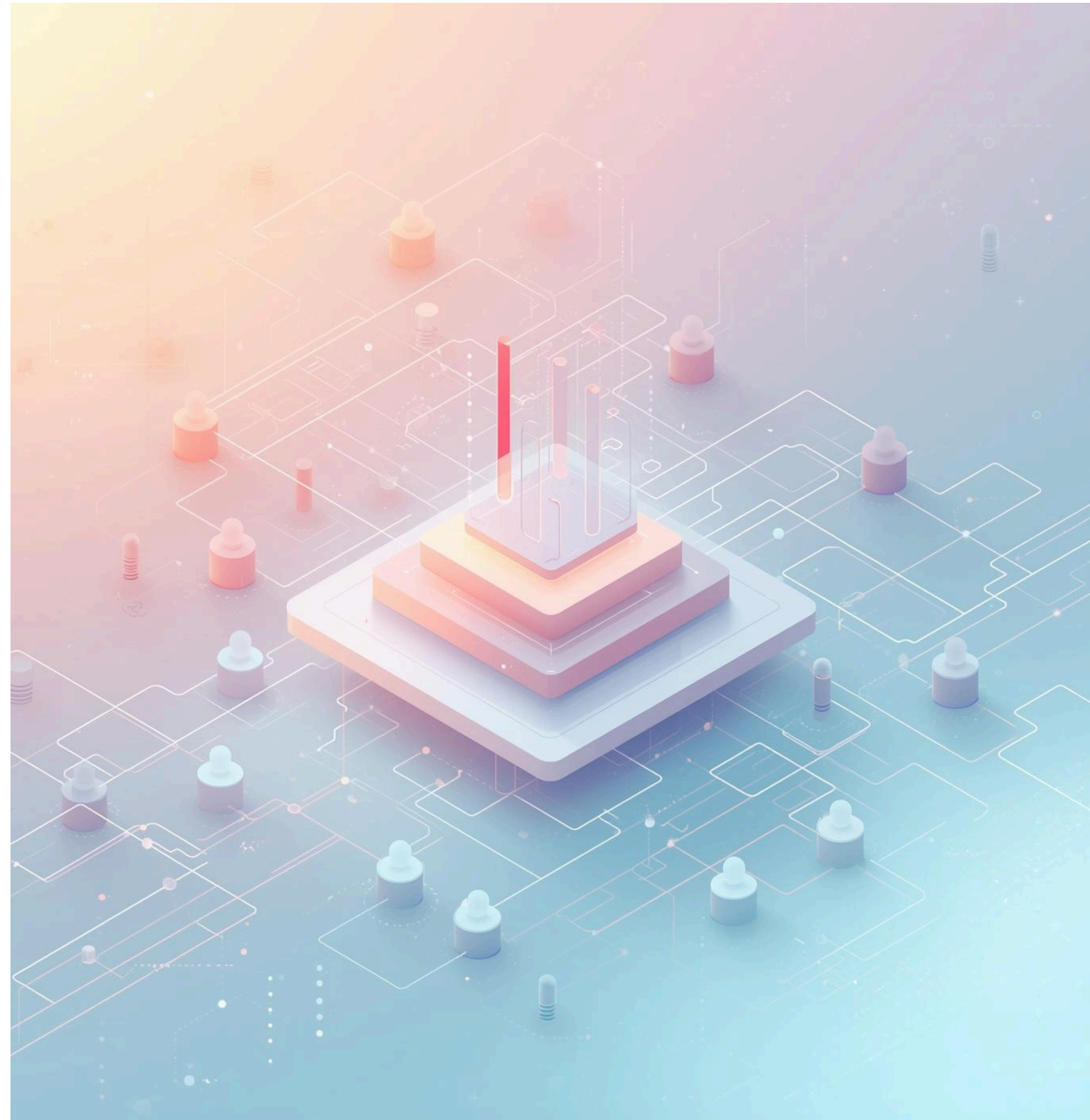
Role of Data Analysis, ML & AI

Current Implementation:

- Rule-based analysis
- Threshold-based decisions
- Trend analysis

AI / ML Integration:

- Supervised learning
 - Uses labelled datasets
 - Models trained using historical data
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AI Decision Capabilities

The system can provide:

- Whether irrigation is needed (YES / NO)
 - Recommended water amount:
 - LOW / MEDIUM / HIGH
 - Seasonal and time-based irrigation guidance
 - Climate and humidity trend analysis
 - Conceptual crop suitability suggestions
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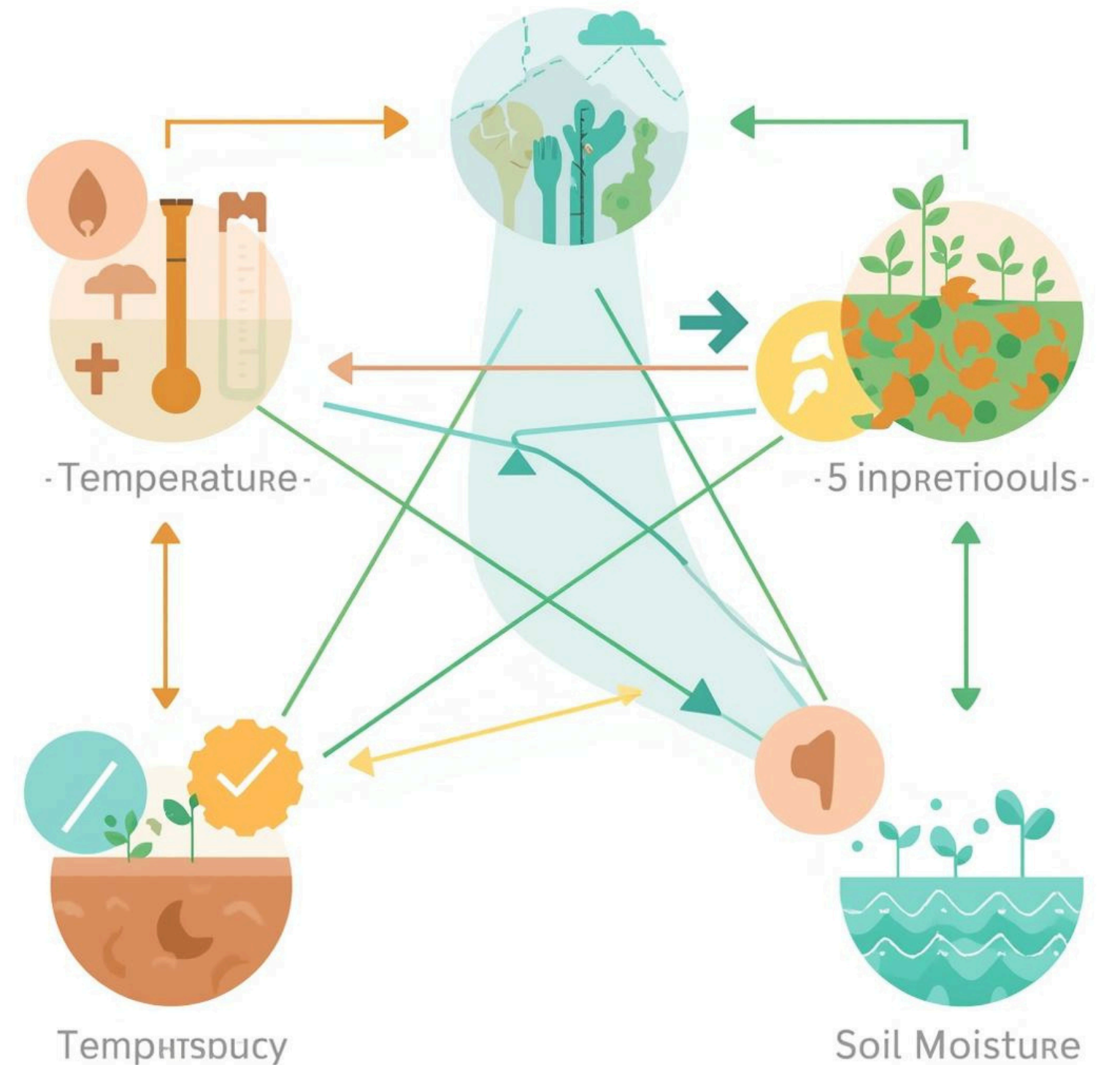


Learning Type & Dataset

- Machine Learning Type: Supervised Learning
- Dataset Type: Labelled data
- Dataset Source: Public datasets (Kaggle)
- Inputs:
 - Temperature
 - Humidity
 - Soil moisture
- Outputs:
 - Irrigation decision
 - Water quantity recommendation

Supervised Learning Model

compares the books of online inputs



Communication Protocols

- Wi-Fi (ESP32)
- HTTP / HTTPS
- REST API
- MQTT (future enhancement)

Reason for selection:

- Low cost
- Easy implementation
- Suitable for academic projects



Network Topology

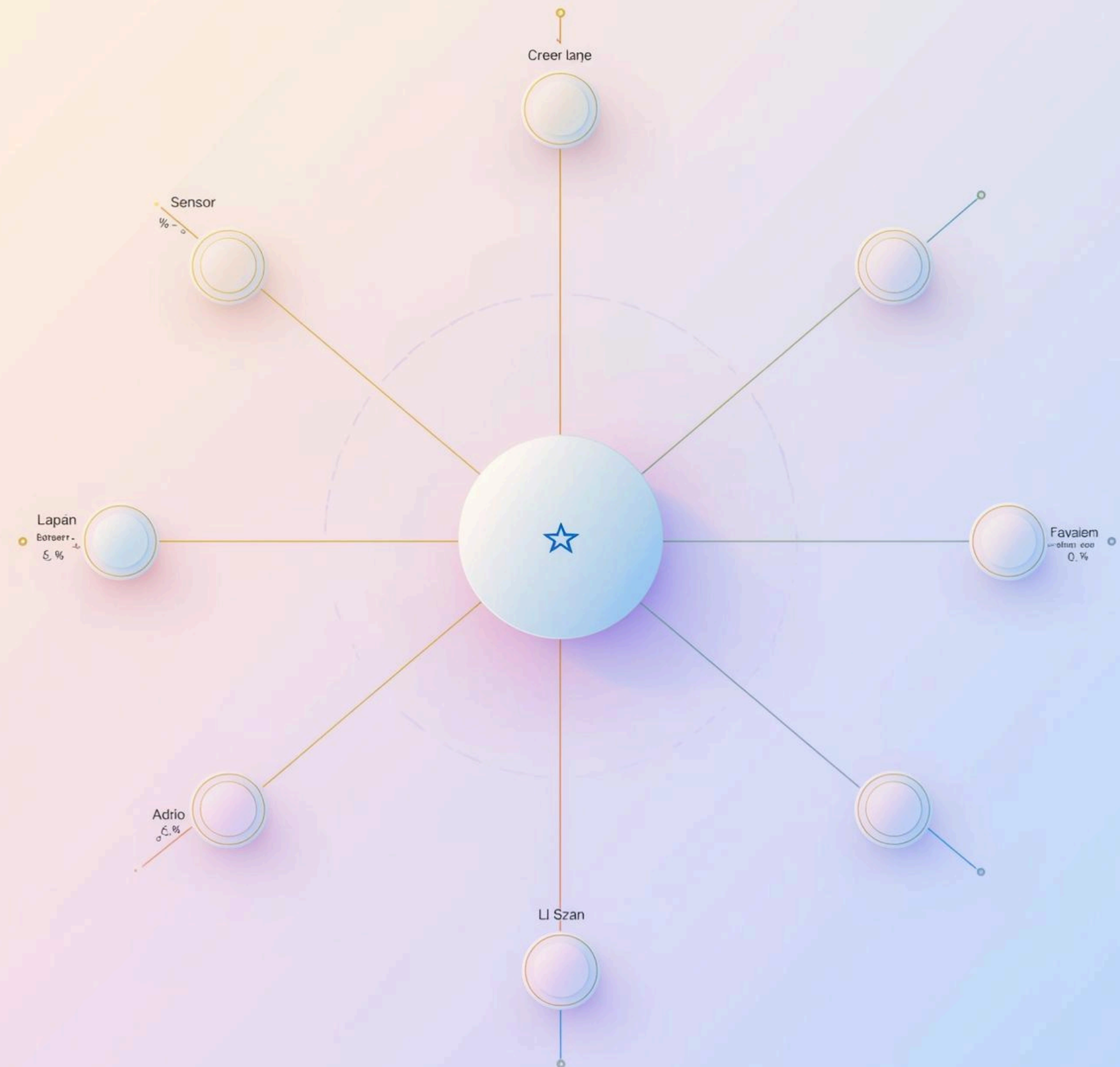
- Topology Type: Star Topology

Structure:

- Sensors → ESP32
- ESP32 → Firebase
- Firebase → Dashboard & Mobile App

Advantages:

- Simple
 - Scalable
 - Reliable
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Expected Outcomes

- Efficient water usage
 - Real-time farm monitoring
 - Data-driven irrigation decisions
 - Climate-aware farming insights
 - Improved agricultural productivity
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Conclusion Overview

- AgroSmart is a complete smart agriculture platform
- Monitoring system is the core component
- Precision system provides intelligent insights
- AI enhances decision-making
- System is scalable and cost-effective



Thank You!!



